

7. Option-Setting Memory

7.1 Overview

The option-setting memory determines the state of the MCU after a reset. The option-setting memory is allocated to the configuration setting area or the OTP area of the extra MRAM memory.

[Figure 7.1](#) shows the option-setting memory area. The option-setting memory area has Secure region and Non-secure region. [Table 7.1](#) shows the programming condition of the option-setting memory area.

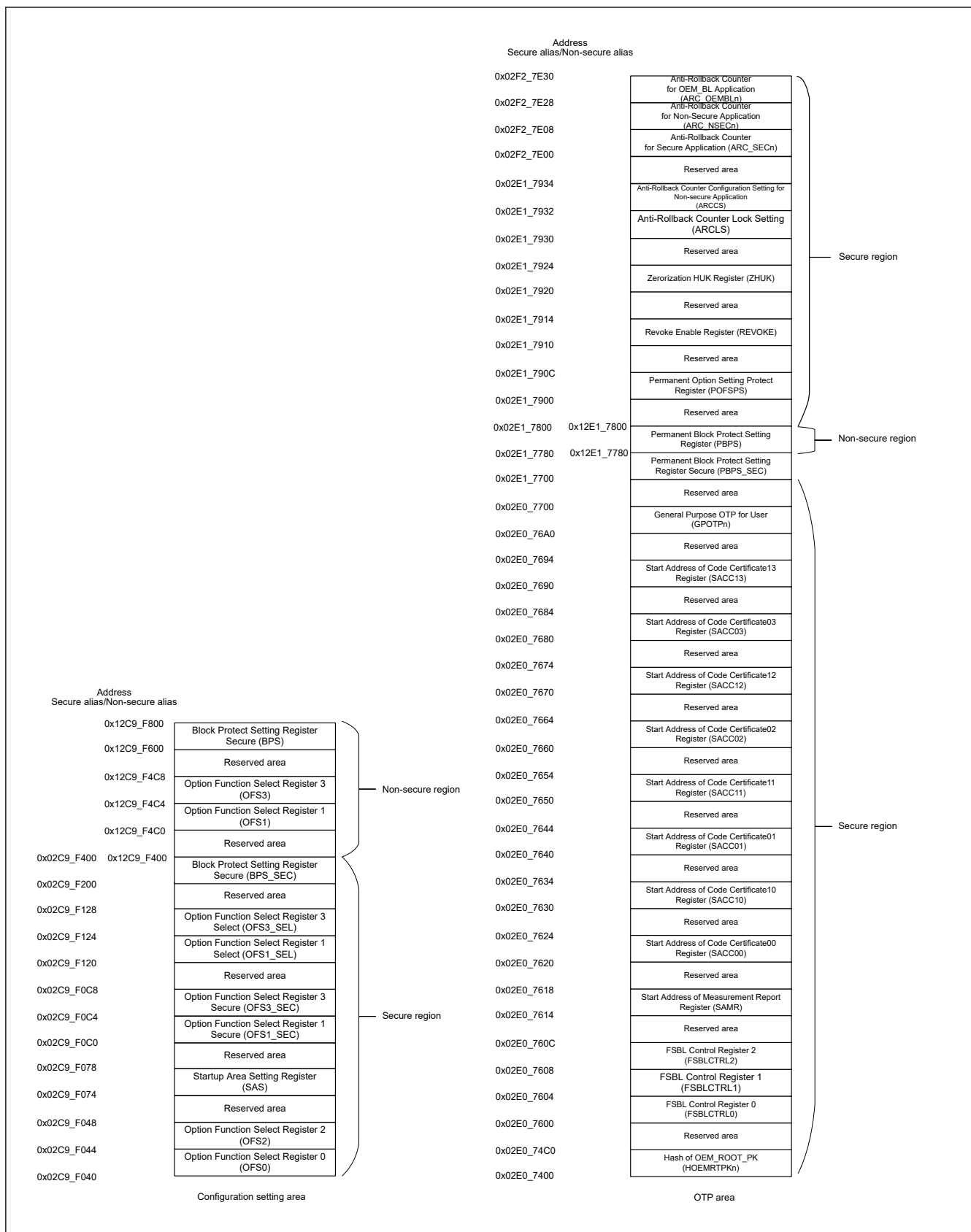


Figure 7.1 Option-setting memory area

Table 7.1 The programming condition of the option-setting memory area

	Self programming	Serial programming	Programming by the on-chip debugger
Secure region	Programming commands issued via Secure alias access.	Programming commands issued when the authentication level is AL2.	Programming commands issued when the authentication level is AL2.
Non-secure region	Programming commands issued via Non-secure alias access.	Programming commands issued when the authentication level is AL2 or AL1.	Programming commands issued when the authentication level is AL2 or AL1.

7.2 Register Descriptions

7.2.1 OFS0 : Option Function Select Register 0

Address: 0x02C9_F040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	WDT0 STPC TL	—	WDT0 RSTIR QS	WDT0RPSS [1:0]	WDT0RPES [1:0]	WDT0CKS[3:0]				WDT0TOPS [1:0]		WDT0 STRT	—		

Value after reset: User setting*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	IWDT STPC TL	—	IWDT RSTIR QS	IWDRPSS[1:0]	IWDRPES[1:0]	IWDTCKS[3:0]				IWDTTOPS[1:0]		IWDT STRT	—		

Value after reset: User setting*1

Bit	Symbol	Function	R/W
0	—	The program value is read from this bit.*2	R
1	IWDTSTRT	IWDT Start Mode Select 0: Automatically activate IWDT after a reset (auto start mode) 1: Stop IWDT after a reset (register start mode)	R
3:2	IWDTTOPS[1:0]	IWDT Timeout Period Select 0 0: 128 cycles (0x007F) 0 1: 512 cycles (0x01FF) 1 0: 1024 cycles (0x03FF) 1 1: 2048 cycles (0x07FF)	R
7:4	IWDTCKS[3:0]	IWDT-Dedicated Clock Frequency Division Ratio Select 0x0: × 1 0x2: × 1/16 0x3: × 1/32 0x4: × 1/64 0xF: × 1/128 0x5: × 1/256 Others: Reserved	R
9:8	IWDRPES[1:0]	IWDT Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (no window end position setting)	R
11:10	IWDRPSS[1:0]	IWDT Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (no window start position setting)	R
12	IWDRSTIRQS	IWDT Reset Interrupt Request Select 0: Interrupt 1: Reset	R
13	—	The program value is read from this bit.*2	R

Bit	Symbol	Function	R/W
14	IWDTSTPCTL	IWDT Stop Control 0: Continue counting 1: Stop counting when primary CPU is in CPU Sleep mode or CPU Deep Sleep mode, or MCU is in Software Standby mode or Deep Software Standby mode 1	R
16:15	—	The program value is read from these bits.*2	R
17	WDT0STRT	WDT0 Start Mode Select 0: Automatically activate WDT0 after a reset (auto start mode) 1: Stop WDT0 after a reset (register start mode)	R
19:18	WDT0TOPS[1:0]	WDT0 Timeout Period Select 0 0: 1024 cycles (0x03FF) 0 1: 4096 cycles (0x0FFF) 1 0: 8192 cycles (0x1FFF) 1 1: 16384 cycles (0x3FFF)	R
23:20	WDT0CKS[3:0]	WDT0 Clock Frequency Division Ratio Select 0x1: PCLKB divided by 4 0x4: PCLKB divided by 64 0xF: PCLKB divided by 128 0x6: PCLKB divided by 512 0x7: PCLKB divided by 2048 0x8: PCLKB divided by 8192 Others: Reserved	R
25:24	WDT0RPES[1:0]	WDT0 Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (no window end position setting)	R
27:26	WDT0RPSS[1:0]	WDT0 Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (no window start position setting)	R
28	WDT0RSTIRQS	WDT0 Reset Interrupt Request Select 0: Interrupt 1: Reset	R
29	—	The program value is read from this bit.*2	R
30	WDT0STPCTL	WDT0 Stop Control 0: Continue counting 1: Stop counting when CPU0 is in CPU Sleep mode or CPU Deep Sleep mode	R
31	—	The program value is read from this bit.*2	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

IWDTSTRT bit (IWDT Start Mode Select)

The IWDTSTRT bit selects the mode in which the IWDT is activated after a reset (stopped state or activated state). When IWDT is activated in auto start mode, the OFS0 register setting for the IWDT is valid.

IWDTTOPS[1:0] bits (IWDT Timeout Period Select)

The IWDTTOPS[1:0] bits specify the timeout period, that is, the time it takes for the down counter to underflow, as 128, 512, 1024, or 2048 cycles of the frequency-divided clock set in the IWDTCKS[3:0] bits. The time it takes for the counter to underflow after a refresh operation is determined by the combination of the IWDTCKS[3:0] and IWDTTOPS[1:0] bits.

For details, see [section 28, Independent Watchdog Timer \(IWDT\)](#).

IWDTCKS[3:0] bits (IWDT-Dedicated Clock Frequency Division Ratio Select)

The IWDTCKS[3:0] bits specify the division ratio of the prescaler for dividing the frequency of the clock for the IWDT as 1/1, 1/16, 1/32, 1/64, 1/128, and 1/256. Using this setting combined with the IWDTTOPS[1:0] bits setting, the IWDT counting period can be set from 128 to 524,288 IWDT clock cycles.

For details, see [section 28, Independent Watchdog Timer \(IWDT\)](#).

IWDTRPES[1:0] bits (IWDT Window End Position Select)

The IWDTRPES[1:0] bits specify the position where the window for the down counter ends as 0%, 25%, 50%, or 75% of the count value. The value of the window end position must be smaller than the value of the window start position, otherwise only the value for the window start position is valid.

The counter values associated with the settings for the start and end positions of the window in the IWDTRPSS[1:0] and IWDTRPES[1:0] bits vary with the setting in the IWDTTOPS[1:0] bits.

For details, see [section 28, Independent Watchdog Timer \(IWDT\)](#).

IWDTRPSS[1:0] bits (IWDT Window Start Position Select)

The IWDTRPSS[1:0] bits specify the position where the window for the down counter starts as 25%, 50%, 75%, or 100% of the counted value. The point at which counting starts is 100% and the point at which an underflow occurs is 0%. The interval between the window starts and ends positions becomes the period in which a refresh is possible. Refresh is not possible outside this period.

For details, see [section 28, Independent Watchdog Timer \(IWDT\)](#).

IWDTRSTIRQS bit (IWDT Reset Interrupt Request Select)

The IWDTRSTIRQS bit selects the operation on an underflow of the down counter or generation of a refresh error. The operation is selectable to an independent watchdog timer reset, a non-maskable interrupt request, or an interrupt request.

For details, see [section 28, Independent Watchdog Timer \(IWDT\)](#).

IWDSTPCTL bit (IWDT Stop Control)

The IWDSTPCTL bit specifies whether to stop counting when entering CPU Sleep mode, CPU Deep Sleep mode, Software Standby mode or Deep Software Standby mode¹.

For details, see [section 28, Independent Watchdog Timer \(IWDT\)](#).

WDT0STRT bit (WDT0 Start Mode Select)

The WDT0STRT bit selects the mode in which the WDT is activated after a reset (stopped state or activated in auto start mode). When WDT is activated in auto start mode, the OFS0 register setting for the WDT is valid.

WDT0TOPS[1:0] bits (WDT0 Timeout Period Select)

The WDT0TOPS[1:0] bits specify the timeout period, that is, the time it takes for the down counter to underflow as 1024, 4096, 8192, or 16384 cycles of the frequency-divided clock set in the WDT0CKS[3:0] bits. The number of PCLKB cycles that takes to underflow after a refresh operation is determined by a combination of the WDT0CKS[3:0] and WDT0TOPS[1:0] bits.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT0CKS[3:0] bits (WDT0 Clock Frequency Division Ratio Select)

The WDT0CKS[3:0] bits specify the division ratio of the prescaler for dividing the frequency of PCLKB as 1/4, 1/64, 1/128, 1/512, 1/2048, and 1/8192. Using this setting combined with the WDT0TOPS[1:0] bits setting, the WDT counting period can be set from 4,096 to 134,217,728 PCLKB cycles.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT0RPES[1:0] bits (WDT0 Window End Position Select)

The WDT0RPES[1:0] bits specify the position where the window on the down counter ends as 0%, 25%, 50%, or 75% of the counted value. The value of the window end position must be smaller than the value of the window start position, otherwise only the value for the window start position is valid.

The counter values associated with the settings for the start and end positions of the window in the WDT0RPSS[1:0] and WDT0RPES[1:0] bits vary with the setting of the WDT0TOPS[1:0] bits.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT0RPSS[1:0] bits (WDT0 Window Start Position Select)

The WDT0RPSS[1:0] bits specify the position where the window for the down counter starts as 25%, 50%, 75%, or 100% of the counted value. The point at which counting starts is 100% and the point at which an underflow occurs is 0%. The interval between the positions where the window starts and ends becomes the period in which a refresh is possible.

Refresh is not possible outside this period.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT0RSTIRQS bit (WDT0 Reset Interrupt Request Select)

The WDT0RSTIRQS bit selects the operation on an underflow of the down-counter or generation of a refresh error. The operation is selectable to a watchdog timer reset, a non-maskable interrupt request, or an interrupt request.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT0STPCTL bit (WDT0 Stop Control)

The WDT0STPCTL bit specifies whether to stop counting when CPU0 is in CPU Sleep mode or CPU Deep Sleep mode.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

7.2.2 OFS2 : Option Function Select Register 2

Address: 0x02C9_F044

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: User setting*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	EXVD DSEL [1:0]	EXVD DSEL [1:0]	—	—	CVMR DIS	DCDC EN

Value after reset: User setting*1

Bit	Symbol	Function	R/W
0	DCDCEN	DCDC enable 0: Disable DCDC 1: Enable DCDC	R
1	CVMRDIS	CVM reset disable 0: Enable CVM reset 1: Disable CVM reset	R
3:2	—	The program value is read from these bits.*2	R
5:4	EXVDDSEL[1:0]	External VDD voltage select for SRAM 1 0: 0.9V ± 10% Others: Setting prohibit	R
31:6	—	The program value is read from these bits.*2	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

DCDCEN bit (DCDC enable)

The DCDCEN bit selects whether the DCDC is enabled or disabled.

CVMRDIS bit (CVM reset disable)

The CVMRDIS bit selects whether the CVM reset is enabled or disabled.

EXVDDSEL[1:0] bits (External VDD voltage select for SRAM)

The EXVDDSEL[1:0] bit selects the voltage for SRAM in external VDD mode.

7.2.3 SAS : Startup Area Setting Register

Address: 0x02C9_F074

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	BTFL G	BTSIZE[1:0]	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: User setting*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: User setting*1

Bit	Symbol	Function	R/W
14:0	—	The program value is read from these bits.*2	R
28:15	—	The program value is read from these bits.*2	R
30:29	BTSIZE[1:0]	Size of Start-Up Area select 0 0: Reserved 0 1 Start-Up area Size is set to 32KB 1 0 Start-Up area Size is set to 16KB 1 1: Start-Up area Size is set to 8KB	R
31	BTFLG	Startup Area Select Flag This bit specifies whether the address of the startup area is exchanged for the boot swap function or not. Example of 8KB startup area 0: First 8 KB area (0x0200_0000 to 0x0200_1FFF) and second 8 KB area (0x0200_2000 to 0x0200_3FFF) are exchanged. 1: First 8 KB area (0x0200_0000 to 0x0200_1FFF) and second 8 KB area (0x0200_2000 to 0x0200_3FFF) are not exchanged.	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

7.2.4 OFS1, OFS1_SEC : Option Function Select Register 1 for Non-Secure and Secure

Address: OFS1: 0x12C9_F4C0
OFS1_SEC: 0x02C9_F0C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	INITE CCEN	SWDB G	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	HOCOFRQ0[2:0]	HOCO EN	—	—	PVDL PSEL	—	PVDA S	—	—	—

Value after reset: The value set by the user*1

Bit	Symbol	Function	R/W
2:0	VDSEL[2:0]	Voltage Detection 0 Level Select 0 0 0: Select 2.85 V 0 0 1: Select 2.58 V 0 1 0: Select 2.15 V 0 1 1: Select 2.00 V 1 0 0: Select 1.90 V 1 0 1: Select 1.80 V 1 1 0: Select 1.67 V 1 1 1: Select 1.56 V*3	R

Bit	Symbol	Function	R/W
3	PVDAS	Voltage Detection 0 Circuit Start 0: Enable voltage monitor 0 reset after a reset 1: Disable voltage monitor 0 reset after a reset	R
4	—	The program value is read from this bit.*2	R
5	PVDLPSEL	Low Power Consumption Function of PVD0 Select at Deep Software standby mode 0: Enable low power consumption function by selecting PVD0 for the battery power supply switch control during DSTBY1 and DSTBY2. 1: Disable low power consumption function of PVD0 for the battery power supply switch control during DSTBY1 and DSTBY2.	R
7:6	—	The program value is read from these bits.*2	R
8	HOCOEN	HOCO Oscillation Enable 0: Enable HOCO oscillation after a reset 1: Disable HOCO oscillation after a reset	R
11:9	HOCOFREQ[2:0]	HOCO Frequency Setting 0 0 0 0: 16 MHz 0 0 1: 18 MHz 0 1 0: 20 MHz 1 0 0: 32 MHz 1 1 1: 48 MHz Others: Reserved.	R
23:12	—	The program value is read from these bits.*2	R
24	SWDBG	Software Debug Control 0: Enable software debug control. The MCU behaves according to DBGCTRL0 and DBGCTRL1, and IWDG and WDT automatically stops when CPU is in the debug state. 1: Disable software debug control. The MCU ignores DBGCTRL0 and DBGCTRL1, and IWDG and WDT continue the operation even when CPU is in the debug state. However, when CDBGPWRUPREQ is 1 and AL is not AL0, IWDG and WDT automatically stops when CPU is in the debug state.	R
25	INITECCEN	Initial ECC Enable 0: Disable ECC function of TCM and CACHE of CPU0. 1: Enable ECC function of TCM and CACHE of CPU0.	R
31:26	—	The program value is read from these bits.*2	R

Note 1. The value of OFS1 and OFS1_SEC in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

Note 3. Setting prohibited when VBATT function is enabled, OFS1(_SEC).PVDAS and PVDLPSEL bits are 0 and at least one of Deep Software Standby modes 1 and 2 is used.

OFS1 register is for Non-secure developer, and OFS1_SEC register is for Secure developer.

VDSEL[2:0] bits (Voltage Detection 0 Level Select)

The VDSEL[2:0] bits select the voltage detection level of the voltage detection 0 circuit.

PVDAS bits (Voltage Detection 0 Circuit Start)

The PVDAS bit selects whether the voltage monitor 0 reset is enabled or disabled after a reset.

PVDLPSEL bits (Low Power Consumption Function of PVD0 Select at Deep Software standby mode)

The PVDLPSEL selects whether the low power consumption function of PVD0 for the battery power supply switch control is enabled or disabled during DSTBY1 and DSTBY2 mode.

When disable this function, the battery power supply switch is controlled by VCC voltage drop detection circuit dedicated for VBATT, the response time is faster instead of increasing current consumption. When enable this function, the battery power supply switch is controlled by PVD0, function, the current consumption is smaller instead of response time delay.

See [section 60, Electrical Characteristics](#)

HOCOEN bit (HOCO Oscillation Enable)

The HOCOEN bit selects whether the HOCO oscillation is enabled or disabled after a reset. Setting this bit to 0 allows the HOCO oscillation to start before the CPU starts operation, which reduces the wait time for oscillation stabilization.

Note: When the HOCOEN bit is set to 0, the system clock source is not switched to HOCO. The system clock source is only switched to HOCO by setting the Clock Source Select bits (SCKSCR.CKSEL[2:0]). To use the HOCO clock, Set the OFS1(_SEC).HOCOFRQ0[2:0] bits to an optimum value.

Note: OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1.HOCOFRQ0[2:0] bits is automatically transferred to HOCOCR2.HCFRQ0[2:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[2:0] bits when OFS1(_SEC).HOCOEN = 1.

HOCOFRQ0[2:0] bits (HOCO Frequency Setting 0)

The HOCOFRQ0[2:0] bits specify the HOCO frequency after a reset as 16 to 48 MHz.

SWDBG bit (Software Debug Control)

The SWDBG bit selects whether software debug control is enabled or disabled.

INITECCEN bit (Initial ECC Enable)

The INITECCEN bit selects whether ECC function of TCM and CACHE of CPU0 is enabled or disabled.

When the INITECCEN bit is changed from 1 to 0, be sure to perform a power-on reset after the change.

7.2.5 OFS1_SEL : Option Function Select Register 1 for Security Attribution

Address: OFS1_SEL: 0x02C9_F120

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	INITECCEN	SWDBG	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	HOCOFRQ0[2:0]	HOCOEN	—	—	PVDLPSEL	—	PVDAS	—	—	VDSEL[2:0]

Value after reset: The value set by the user*1

Bit	Symbol	Function	R/W
2:0	VDSEL[2:0]	Security attributes of Voltage Detection 0 Level Select 0 0 0: Select OFS1_SEC.VDSEL0[2:0] 1 1 1: Select OFS1.VDSEL0[2:0] Others: Select Reserved.	R
3	PVDAS	Security attributes of Voltage Detection 0 Circuit Start 0: Select OFS1_SEC.PVDAS 1: Select OFS1.PVDAS	R
4	—	The program value is read from this bit.*2	R
5	PVDLPSEL	Security attributes of Low Power Consumption Function of PVD0 Select at Deep Software standby mode 0: Select OFS1_SEC.PVDLPSEL 1: Select OFS1.PVDLPSEL.	R
7:6	—	The program value is read from these bits.*2	R
8	HOCOEN	Security attributes of HOCO Oscillation Enable 0: Select OFS1_SEC.HOCOEN 1: Select OFS1.HOCOEN	R

Bit	Symbol	Function	R/W
11:9	HOCOFREQ[2:0]	Security attributes of HOCO Frequency Setting 0 0 0 0: Select OFS1_SEC.HOCOFREQ[2:0] 1 1 1: Select OFS1.HOCOFREQ[2:0] Others: Reserved	R
23:12	—	The program value is read from these bits.*2	R
24	SWDBG	Security attributes of Software Debug Control 0: Select OFS1_SEC.SWDBG 1: Select OFS1.SWDBG	R
25	INITECCEN	Security attributes of Initial ECC Enable 0: Select OFS1_SEC.INITECCEN 1: Select OFS1.INITECCEN	R
31:26	—	The program value is read from these bits.*2	R

Note 1. The value of OFS1_SEL in a blank product is 0x00000000. It is set to the value written by your application

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 0.

The OFS1_SEL register selects the security attribute. Which security attribute is applied, OFS or OFS_SEC, is determined by the setting value of the corresponding bit in the OFS1_SEL register. For details, see [section 7.3.3. Security Attribution of Extra MRAM Option-Setting Memory](#)

7.2.6 OFS3, OFS3_SEC : Option Function Select Register 3

Address: OFS3: 0x12C9_F4C4
OFS3_SEC: 02C9_F0C4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	WDT1 STPC TL	—	WDT1 RSTIR QS	WDT1RPSS [1:0]	WDT1RPES [1:0]	WDT1CKS[3:0]			WDT1TOPS [1:0]	WDT1 STRT	—				

Value after reset: User setting*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: User setting*1

Bit	Symbol	Function	R/W
16:0	—	The program value is read from these bits. *2	R
17	WDT1STRT	WDT1 Start Mode Select 0: Automatically activate WDT after a reset (auto start mode) 1: Stop WDT after a reset (register start mode).	R
19:18	WDT1TOPS[1:0]	WDT1 Timeout Period Select 0 0: 1024 cycles (0x03FF) 0 1: 4096 cycles (0x0FFF) 1 0: 8192 cycles (0x1FFF) 1 1: 16384 cycles (0x3FFF)	R
23:20	WDT1CKS[3:0]	WDT1 Clock Frequency Division Ratio Select 0 0 0 1: PCLKB divided by 4 0 1 0 0: PCLKB divided by 64 1 1 1 1: PCLKB divided by 128 0 1 1 0: PCLKB divided by 512 0 1 1 1: PCLKB divided by 2048 1 0 0 0: PCLKB divided by 8192 Other settings are prohibited.	R

Bit	Symbol	Function	R/W
25:24	WDT1RPES[1:0]	WDT1 Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (No window end position setting)	R
27:26	WDT1RPSS[1:0]	WDT1 Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (No window start position setting)	R
28	WDT1RSTIRQS	WDT1 Reset Interrupt Request Select WDT Behavior Select 0: NMI 1: Reset	R
29	—	The program value is read from these bits. *2	R
30	WDT1STPCTL	WDT1 Stop Control 0: Continue counting 1: Stop counting when CPU1 is in CPU Sleep mode or CPU Deep Sleep mode	R
31	—	The program value is read from these bits. *2	R

Note 1. The value of OFS3 and OFS3_SEC in a blank product is 0xFFFFFFFF. The value of OFS3_SEL in factory shipment product is 0x00000000. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 0.

OFS3 register is for Non-secure developer, and OFS3_SEC register is for Secure developer. OFS3 register is for non-secure developer.

WDT1STRT bit (WDT1 Start Mode Select)

The WDT1STRT bit selects the mode in which the WDT is activated after a reset (stopped state or activated state). When the WDT is activated in auto start mode, the OFS0 register setting for the WDT is valid.

WDT1TOPS[1:0] bits (WDT1 Timeout Period Select)

The WDT1TOPS[1:0] bits select the timeout period, the time it takes for the down counter to underflow, as 1024, 4096, 8192, or 16384 cycles of the frequency-divided clock set in the WDT1CKS[3:0] bits. The number of PCLKB cycles that the counter takes to underflow after a refresh operation is determined by the combination of the WDT1CKS[3:0] and WDT1TOPS[1:0] bits.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT1CKS[3:0] bits (WDT1 Clock Frequency Division Ratio Select)

The WDT1CKS[3:0] bits select the division ratio of the prescaler to divide the PCLKB frequency as 1/4, 1/64, 1/128, 1/512, 1/2048, or 1/8192. Using this setting combined with the WDT1TOPS[1:0] bit setting, the WDT counting period can be set from 4,096 to 134,217,728 PCLKB cycles.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT1RPES[1:0] bits (WDT1 Window End Position Select)

The WDT1RPES[1:0] bits select the position where the window for the down counter ends as 0%, 25%, 50%, or 75% of the counted value. The value of the window end position must be smaller than the value of the window start position. Otherwise, only the value for the window start position is valid.

The counter values associated with the settings for the start and end positions of the window, in the WDT1RPSS[1:0] and WDT1RPES[1:0] bits, vary with the setting of the WDT1TOPS[1:0] bits.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT1RPSS[1:0] bits (WDT1 Window Start Position Select)

The WDT1RPSS[1:0] bits select the position where the window for the down counter starts as 25%, 50%, 75%, or 100% of the counted value. The point at which counting starts is 100% and the point at which an underflow occurs is 0%.

The interval between the positions where the window starts and ends becomes the period in which a refresh is possible. However, refresh is not possible outside this period.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT1RSTIRQS bit (WDT1 Reset Interrupt Request Select)

The WDT1RSTIRQS bit selects the operation on an underflow of the down counter or generation of a refresh error. The selected operation can be a watchdog timer reset, non-maskable interrupt request, or interrupt request.

For details, see [section 27, Watchdog Timer \(WDT\)](#).

WDT1STPCTL bit (WDT1 Stop Control)

The WDT1STPCTL bit specifies whether to stop counting when CPU1 is in CPU Sleep mode or CPU Deep Sleep mode. For details, see [section 27, Watchdog Timer \(WDT\)](#).

7.2.7 OFS3_SEL : Option Function Select Register 3 for Security

Address: OFS3_SEL: 02C9 F124h

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	WDT1 STPC TL	—	WDT1 RSTIR QS	WDT1RPSS [1:0]	WDT1RPES [1:0]	WDT1CKS[3:0]				WDT1TOPS [1:0]	WDT1 STRT	—			
Value after reset:	User setting*1															
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	User setting*1															

Bit	Symbol	Function	R/W
16:0	—	The program value is read from these bits. *2	R
17	WDT1STRT	Security attributes of WDT1 Start Mode Select 0: Select OFS3_SEC.WDT1STRT 1: Select OFS3.WDT1STRT.	R
19:18	WDT1TOPS[1:0]	Security attributes of WDT1 Timeout Period Select 0 0: Select OFS3_SEC.WDT1TOPS[1:0] 1 1: Select OFS3.WDT1TOPS[1:0] Others: Setting prohibit	R
23:20	WDT1CKS[3:0]	Security attributes of WDT1 Clock Frequency Division Ration Select 0 0 0 0: Select OFS3_SEC.WDT1CKS[3:0] 1 1 1 1: Select OFS3.WDT1CKS[3:0] Others: Setting prohibit	R
25:24	WDT1RPES[1:0]	Security attributes of WDT1 Window End Position Select 0 0: Select OFS3_SEC.WDT1RPES[1:0] 1 1: Select OFS3.WDT1RPES[1:0] Others: Setting prohibit	R
27:26	WDT1RPSS[1:0]	Security attributes of WDT1 Window Start Position Select 0 0: Select OFS3_SEC.WDT1RPSS[1:0] 1 1: Select OFS3.WDT1RPSS[1:0] Others: Setting prohibit	R
28	WDT1RSTIRQS	Security attributes of WDT1 Reset Interrupt Request Select 0: Select OFS3_SEC.WDT1RSTIRQS 1: Select OFS3.WDT1RSTIRQS	R
29	—	The program value is read from these bits. *2	R

Bit	Symbol	Function	R/W
30	WDT1STPCTL	Security attributes of WDT1 Stop Control 0: Select OFS3_SEC.WDT1STPCTL 1: Select OFS3.WDT1STPCTL	R
31	—	The program value is read from these bits. *2	R

Note 1. The value of OFS3 and OFS3_SEC in a blank product is 0xFFFFFFFF. The value of OFS3_SEL in factory shipment product is 0x00000000. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 0.

The OFS3_SEL register selects the security attribute. Which security attribute is applied, OFS or OFS_SEC, is determined by the setting value of the corresponding bit in the OFS3_SEL register. For details, see [section 58, MRAM](#)

7.2.8 BPS, BPS_SEC : Block Protect Setting Register

Address:

BPS: 0x12C9_F600 + 0x04 × n (n = 0 to 127) (Non-secure)
BPS_SEC: 0x02C9_F200 + 0x04 × n (n = 0 to 127) (Secure)

Bit position: 31

0

Bit field:

Value after reset:

User setting*1

Note 1. The value of BPS and BPS_SEC in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

BPS_SEC register is for Secure developer, and BPS register is for Non-secure developer.

The BPS and BPS_SEC registers invalidate the writing to the code MRAM memory. When the bit of this register is set to 0, the writing to the corresponding block are invalid. For details, see [section 58, MRAM](#).

7.2.9 PBPS, PBPS_SEC : Permanent Block Protect Setting Register

Address: PBPS: 0x12E1_7780 + 0x04 × n (n = 0 to 31) (Non-Secure)
PBPS_SEC: 0x02E1_7700 + 0x04 × n (n = 0 to 31) (Secure)

Bit position: 31

0

Bit field:

Value after reset:

User setting*1

Note 1. The value of PBPS and PBPS_SEC in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

PBPS_SEC register is for secure developer, and PBPS register is for non-secure developer. The PBPS and PBPS_SEC registers invalidate writes to bits of BPS and BPS_SEC. The bit of this register can be programmed 0 when corresponding bit of BPS and BPS_SEC is programmed to 0. When the bit of this register is programmed to 0, writing the corresponding bit of BPS and BPS_SEC register is invalid. Once the bit of this register is set to 0, it is impossible to change the bit to 1.

For details, see [section 58, MRAM](#).

7.2.10 Permanent Option Function Select Protect Setting Register (POFSPS)

Address: 0x02E1_7900

Bit position: 31

0

Bit field:

Value after reset:

User setting*1

Address: 0x02E1_7904

Bit position: 63

32

Bit field: POFSPS[63:32]

Value after reset:

User setting^{*1}

Address: 0x02E1_7908

Bit position: 95

80 79

64

Bit field: Reserved bits POFSPS[79:64]

Value after reset:

User setting^{*1}

Note 1. The value of POFSPS in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

The POFSPS register invalidate the programming and erasure to the option function select area. When the bit of this register is set to 0, the programming and erasure to the corresponding area is invalid. For details, see [section 58, MRAM](#).

7.2.11 FSBLCTRL0 : FSBL Control Register 0

Address: 0x02E0_7600

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field: — — — — — — — — — — — — — — —

Value after reset:

User setting^{*1}

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: — — — — FSBLCLK[2:0] FSBLSKIPDS[2:0] FSBLSKIPSW[2:0] FSBLLEN[2:0]

Value after reset:

User setting^{*1}

Bit	Symbol	Function	R/W
2:0	FSBLLEN[2:0]	FSBL Enable 0 0 0: Enable FSBL 1 1 1: Disable FSBL Others: Reserved	R
5:3	FSBLSKIPSW[2:0]	FSBL Skip Enable for Software Reset 0 0 0: Enable skip 1 1 1: Disable skip Others: Reserved	R
8:6	FSBLSKIPDS[2:0]	FSBL Skip Enable for Deep Software Standby Reset 0 0 0: Enable skip 1 1 1: Disable skip Others: Reserved	R
11:9	FSBLCLK[2:0]	Clock Frequency Selection during FSBL Execution See Table 7.2 .	R
31:12	—	The program value is read from these bits. ^{*2}	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

Table 7.2 The relationship between FSBLCLK setting and system clock frequency (1 of 2)

System clock	FSBLCLK[2:0] = 111b (VDD = VSCR_1 or voltage range 1)	FSBLCLK[2:0] = 110b (VDD = VSCR_1 or voltage range 1)	FSBLCLK[2:0] = 101b (VDD = VSCR_2 or voltage range 2)	FSBLCLK[2:0] = 100b (VDD = VSCR_2 or voltage range 2)
CPUCLK0	250 MHz	200 MHz	150 MHz	133 MHz

Table 7.2 The relationship between FSBLCLK setting and system clock frequency (2 of 2)

System clock	FSBLCLK[2:0] = 111b (VDD = VSCR_1 or voltage range 1)	FSBLCLK[2:0] = 110b (VDD = VSCR_1 or voltage range 1)	FSBLCLK[2:0] = 101b (VDD = VSCR_2 or voltage range 2)	FSBLCLK[2:0] = 100b (VDD = VSCR_2 or voltage range 2)
CPUCLK1	250 MHz	200 MHz	150 MHz	133 MHz
ICLK	250 MHz	200 MHz	150 MHz	133 MHz
MRCLK	250 MHz	200 MHz	150 MHz	133 MHz
MRPCLK	125 MHz	100 MHz	75 MHz	66 MHz
PCLKA	125 MHz	100 MHz	75 MHz	66 MHz
PCLKB	62.5 MHz	50 MHz	37.5 MHz	33 MHz

Note: Other FSBLCLK settings are prohibited.

7.2.12 FSBLCTRL1 : FSBL Control Register 1

Address: 0x02E0_7604

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: User setting^{*1}

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FSBLEXMD[1:0]	—

Value after reset: User setting^{*1}

Bit	Symbol	Function	R/W
1:0	FSBLEXMD[1:0]	FSBL Execution Mode 0 0: CRC boot and not report measurement 0 1: CRC boot and report measurement 1 0: Secure boot and not report measurement 1 1: Secure boot and report measurement	R
31:2	—	The program value is read from these bits. ^{*2}	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

7.2.13 FSBLCTRL2 : FSBL Control Register 2

Address: 0x02E0_7608

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: User setting^{*1}

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PORTGN[4:0]					PORTPN[3:0]			

Value after reset: User setting^{*1}

Bit	Symbol	Function	R/W
3:0	PORTPN[3:0]	FSBL Error Notification Port Pin Number* ² 0 0 0 0: PORTn00 0 0 0 1: PORTn01 ⋮ 1 1 1 1: PORTn15 (n = 0 to 9, A, B)	R
8:4	PORTGN[4:0]	FSBL Error Notification Port Group Name* ² 0 0 0 0 0: PORT0m 0 0 0 0 1: PORT1m ⋮ 0 1 0 1 1: PORTBm Others: Reserved (m = 0 to 15)	R
31:9	—	The program value is read from these bits.* ³	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Note 2. When non-existent pin is selected, FSBL error notification is not output.

Note 3. This register can only be programmed by MACI command. When programming, the set value should be 1.

FSBLCTRL2 register selects the port to output when an Error occurs during FSBL execution. The PORTGN bits select a group of ports, and the PORTPN bits select the pins of that group. It thereby determines the unique port.

7.2.14 SACC0n : Start Address of Code Certificate 0n (n = 0 to 3)

Address: SACC0n: 0x02E0_7620 + 0x20 × n (n = 0 to 3)

Bit position: 31 0

Bit field:

Value after reset:

User setting*¹

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

See [section 50.7. Secure Boot](#) for details.

7.2.15 SACC1n : Start Address of Code Certificate 1n (n= 0 to 3)

Address: SACC1n: 0x02E0_7630 + 0x20 × n (n = 0 to 3)

Bit position: 31 0

Bit field:

Value after reset:

User setting*¹

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

See [section 50.7. Secure Boot](#) for details.

7.2.16 SAMR : Start Address of Measurement Report

Address: 0x02E0_7614

Bit position: 31 0

Bit field:

Value after reset:

User setting*¹

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

The SAMR register specifies the starting address of measurement report. The measurement report is stored in the SRAM shown in . Specify the address of SRAM0 to SRAM3. In case SRAM0 is used for measurement report, do not specify the

addresses from 0x22007780 to 0x22007BFF. When reading the measurement report, disable the ECC function by setting SRAMCRn.ECCMOD[1:0] = 00b. See [section 50.7. Secure Boot](#) for details.

7.2.17 REVOKE : REVOKE Enable

Address: 0x02E1_7910

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	User setting* ¹															
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	REVOKE[3:0]			
Value after reset:	User setting* ¹															

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Bit	Symbol	Function	R/W
3:0	REVOKE[3:0]	Configuration setting for OEM_ROOT_KEY revoke xxx0b: OEM_ROOT_PK0 is revoked, and read data is masked to 0 xxx1b: OEM_ROOT_PK0 is enabled and can be read xx0xb: OEM_ROOT_PK1 is revoked, and read data is masked to 0 xx1xb: OEM_ROOT_PK1 is enabled and can be read x0xxb: OEM_ROOT_PK2 is revoked, and read data is masked to 0 x1xxb: OEM_ROOT_PK2 is enabled and can be read 0xxxb: OEM_ROOT_PK3 is revoked, and read data is masked to 0 1xxxb: OEM_ROOT_PK3 is enabled and can be read	R
31:4	—	The program value is read from this bit. ^{*1}	R

Note 1. This register can only be programmed by MACI command. When programming, the set value should be 1.

7.2.18 HOEMRTRKn : Hash of OEM_ROOT_PKn (n= 0 to 3)

Address: HOEMRTPK0: 0x02E0_7400 +0x4 × i (i = 0 to 11)
HOEMRTPK1: 0x02E0_7430 +0x4 × i (i = 0 to 11)
HOEMRTPK2: 0x02E0_7460 +0x4 × i (i = 0 to 11)
HOEMRTPK3: 0x02E0_7490 +0x4 × i (i = 0 to 11)

Bit position:	31	0
Bit field:		
Value after reset:	User setting ^{*1}	

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

This register can be programmed only by boot firmware.

7.2.19 ZHUK : Zeroization HUK Enable

Address: 0x02E1_7920

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	User setting*1															
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	ZHUK EN3	ZHUK EN2	ZHUK EN1	ZHUK EN0
Value after reset:	User setting*1															

Bit	Symbol	Function	R/W
0	ZHUKEN0	Configuration setting for RTCIC0 0: W-HUK zeroization by RTCIC0 tamper event is enabled 1: W-HUK zeroization by RTCIC0 tamper event is disabled	R
1	ZHUKEN1	Configuration setting for RTCIC1 0: W-HUK zeroization by RTCIC1 tamper event is enabled 1: W-HUK zeroization by RTCIC1 tamper event is disabled	R
2	ZHUKEN2	Configuration setting for RTCIC2 0: W-HUK zeroization by RTCIC2 tamper event is enabled 1: W-HUK zeroization by RTCIC2 tamper event is disabled	R
3	ZHUKEN3	Configuration setting for MREZC register 0: W-HUK zeroization by MREZC is enabled 1: W-HUK zeroization by MREZC is disabled	R
31:4	—	The program value is read from this bit.*2	R

Note: When ZHUKEN3 bit is 0, set ZHUKENn (n = 0 to 2) bits setting same as VBTADCR3 (see [section 12, Battery Backup Function](#)).
For example, when ZHUKENn (n = 0 to 2) is 101b, set VBTADCR3.VBATADZEN (n = 0 to 2) to 010b.

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

7.2.20 ARCLS : Anti-Rollback Counter Lock Setting

Address: 0x02E1_7930

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	ARCB L_LK	ARCNS_LK[3:0]			ARCS _LK	
Value after reset:	User setting*1															

Bit	Symbol	Function	R/W
0	ARCS_LK	ARC_SEC Lock 0: ARC_SEC protection is enabled 1: ARC_SEC protection is disabled	R
4:1	ARCNS_LK[3:0]	ARC_NSEC Lock In case of 64 bits x 4 counter setting x x x 0: ARC_NSEC[63:0] protection is enabled x x 0 x: ARC_NSEC[127:64] protection is enabled x 0 x x: ARC_NSEC[191:128] protection is enabled 0 x x x: ARC_NSEC[255:192] protection is enabled 1 1 1 1: ARC_NSEC[255:0] protection is disabled Others Reserved. In case of 256 bits counter setting 0 0 0 0: ARC_NSEC[255:0] protection is enabled 1 1 1 1: ARC_NSEC[255:0] protection is enabled Others Reserved.	R

Bit	Symbol	Function	R/W
5	ARCBL_LK	ARC_OENBL Lock 0: ARC_OEMBL protection is enabled 1: ARC_OEMBL protection is disabled	R
15:6	—	The program value is read from these bits*2	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

Anti-rollback counter Lock bits setting are located in the OTP area. Anti-rollback counter Lock bits setting can be set to "0" by the "Program" command and cannot be changed from "0" to "1". When the lock bit is 0, the anti-rollback counter corresponding to each bit cannot be updated by the "Increment Counter" command.

7.2.21 ARCCS : Anti-Rollback Counter Configuration Setting for Non-secure Application

Address: 0x02E1_7932

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CNF_ARCNS [1:0]

Value after reset:

User setting*1

Bit	Symbol	Function	R/W
1:0	CNF_ARCNS[1:0]	Configuration setting for ARC_NSEC 0 0: ARC_NSEC configuration is 4 × 64 bits 0 1: ARC_NSEC configuration is 1 × 256 bits Others ARC_NSEC configuration is disabled	R
15:2	—	The program value is read from these bits*2	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application

Note 2. This register can only be programmed by MACI command. When programming, the set value should be 1.

ARC_NSEC configuration setting is located in the OTP area. These bits specify two types of ARC_NSEC configuration: 4 × 64 bits or 1 × 256 bits. "Increment Counter" or "Read Counter" command cannot be issued to ARC_NSEC when counter configuration setting is disabled.

7.2.22 ARC_SEcn : Anti-Rollback Counter for Secure Application n (n = 0, 1)

Address: 0x02F2 7E00 + 4 × n (n=0, 1)

Bit position:	31	0
Bit field:	ARC_SEC[32 × n + 31 : 32 × n]	

Value after reset:

User setting*1

Bit	Symbol	Function	R/W
31:0	ARC_SEC[32 × n + 31 : 32 × n]	Anti-Rollback Counter for Secure Application The counter value is obtained by arranging the read values from the upper register (n = 1) to the lower register (n = 0).	R

Note 1. The value in a blank product is 0x0. It is set to the value written by your application

7.2.23 ARC_NSECn : Anti-Rollback Counter for Non-secure Application n (n = 0 to 7)

Address: $0x02F2\ 7E08 + 4 \times n$ (n = 0 to 7)

Bit position: 31 0

Bit field: $ARC_NSECn[32 \times n + 31 : 32 \times n]$

Value after reset: User setting^{*1}

Bit	Symbol	Function	R/W
31:0	$ARC_NSECn[32 \times n + 31 : 32 \times n]$	Anti-Rollback Counter for Non-secure Application The counter value is obtained by arranging the read values from the upper register (n = 7) to the lower register (n = 0).	R

Note 1. The value in a blank product is 0x0. It is set to the value written by your application

7.2.24 ARC_OEMBLn : Anti-Rollback Counter for OEMBL (n = 0, 1)

Address: $0x02F2\ 7E28 + 4 \times n$ (n = 0, 1)

Bit position: 31 0

Bit field: $ARC_OEMBL[32 \times n + 31 : 32 \times n]$

Value after reset: User setting^{*1}

Bit	Symbol	Function	R/W
31:0	$ARC_OEMBL[32 \times n + 31 : 32 \times n]$	Anti-Rollback Counter for OEM_BL Application The counter value is obtained by arranging the read values from the upper register (n = 1) to the lower register (n = 0).	R

Note 1. The value in a blank product is 0x0. It is set to the value written by your application

7.2.25 GPOTPn : General Purpose OTP (n = 0 to 23)

Address: $02E0\ 76A0h + 4 \times n$ (n = 0 to 23)

Bit position: 31 0

Bit field: GPOTP[31:0]

Value after reset: User setting^{*1}

Bit	Symbol	Function	R/W
31:0	GPOTP[31:0]	General Purpose OTP This is an OTP area that can be used for general purpose. Secure user can write any value.	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

7.3 Setting Option-Setting Memory

7.3.1 Allocation of Data in Option-Setting Memory

Programming data is allocated to the addresses in the option-setting memory shown in [Figure 7.1](#). The allocated data is used by FSBL or tools such as a flash programming software or an on-chip debugger.

Note: Programming formats vary depending on the compiler. See the compiler manual for details.

7.3.2 Setting Data for Programming Option-Setting Memory

Allocating data according to the procedure described in [section 7.3.1. Allocation of Data in Option-Setting Memory](#), alone does not actually write the data to the option-setting memory. You must also follow one of the actions described in this section.

(1) Changing the option-setting memory by self-programming

Use the configuration setting command to write data to the option-setting memory in the configuration setting area.

The extra MRAM option-setting memory support background operations (BGO). When write the option-setting memory, the code MRAM can be read.

For details of the configuration setting command, see [section 58, MRAM](#).

(2) Debugging through an OCD or programming by a flash writer

This procedure depends on the tool in use, see the tool manual for details.

The MCU provides two setting procedures:

- Read the data allocated as described in [section 7.3.1. Allocation of Data in Option-Setting Memory](#), from an object file or Motorola S-format file generated by the compiler, and write the data to the MCU
- Use the GUI interface of the tool to program the same data as allocated in [section 7.3.1. Allocation of Data in Option-Setting Memory](#).

7.3.3 Security Attribution of Extra MRAM Option-Setting Memory

Some functionality has 3 registers for Non-secure (FUNC NAME), and Secure (FUNC NAME_SEC), and security attribution (FUNC NAME_SEL). Only Secure developer can set the registers for Secure and security attribution. As shown in [Figure 7.2](#), when the bit of security attribution register is set to 0, the corresponding bit of Secure register is applied. When the bit of security attribution register is set to 1, the corresponding bit of Non-secure register is applied.

For example, if the Secure developer wants to configure PVD of OFS1 as Secure, HOCO of OFS1 as Non-secure, the Secure developer needs to set OFS1_SEL as follows.

OFS1_SEL = 0xFFFF_FFF0

By this setting, PVDAS and VDSEL[2:0] values of OFS1_SEC and HOCOFREQ0[2:0] and HOCOEN values of OFS1 are applied to MCU. The reserved bits of the security attribution register (FUNC NAME_SEL) should be set to 0.

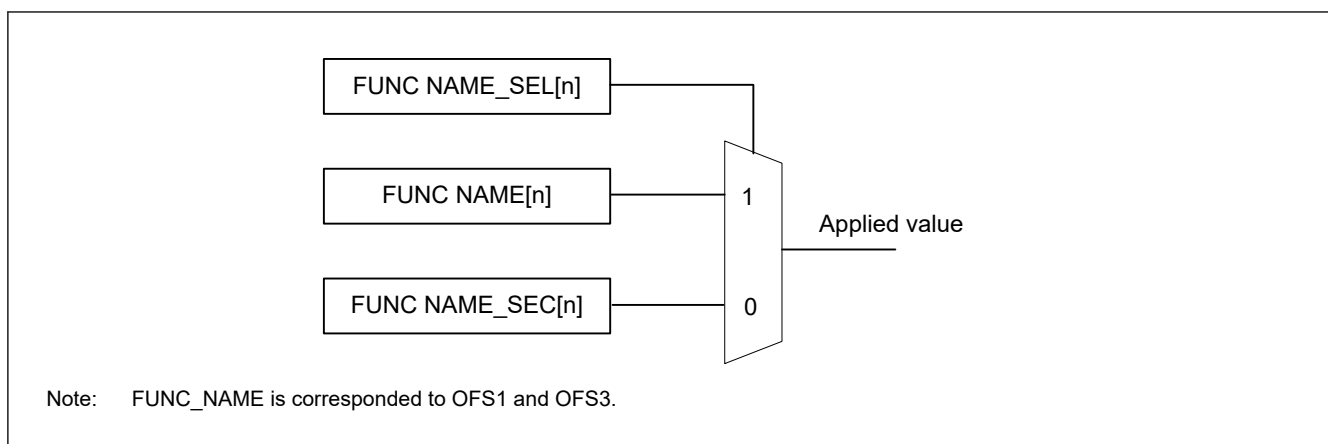


Figure 7.2 Selection of applied value

Block protect setting (BPS or BPS_SEC) and permanent block protect setting (PBPS or PBPS_SEC) are selected by the code MRAM security attribution.

7.3.4 Timing of the Setting Value

For SAS, BPS, BPS_SEC, PBPS and PBPS_SEC registers, the setting value of the related startup area and block protection is applied immediately after programming. For other registers, the setting value is applied after the MCU is reset.

7.4 Usage Notes

7.4.1 Data for Programming Reserved Areas and Reserved Bits in the Option-Setting Memory

- Security attribution region (0x02C9_F120 to 0x02C9_F1FF)
When reserved areas and reserved bits in the option-setting memory are within the scope of programming, write 0 to all bits in reserved areas and all reserved bits. Operation is not guaranteed if 1 is written to these bits.
- Other region
When reserved areas and reserved bits in the option-setting memory are within the scope of programming, write 1 to all bits of reserved areas and all reserved bits. If 0 is written to these bits, normal operation cannot be guaranteed.

7.4.2 FSBL Execution After System Reset

When FSBLCTRL0.FSBLEN bit is set to 000b, FSBL is always executed after system reset is released. However, setting FSBLCTRL0.FBLSKIPSW or FSBLCTRL0.FBLSKIPDS bit adds a condition to Skip FSBL. See [Table 7.3](#).

Table 7.3 FSBL execution conditions

FSBLEN	FBLSKIPSW	FBLSKIPDS	FSBL execution after system reset
111b	*	*	Not executed
000b	111b	111b	Executed
000b	111b	000b	Deep Software Standby reset : Not executed Other reset than above : Executed
000b	000b	111b	Software reset : Not executed Other reset than above : Executed
000b	000b	000b	Software reset : Not executed Deep Software Standby reset : Not executed Other reset than above : Executed

When using the FSBL skip function, the reset status registers (RSTSR0, RSTSR1, and RSTSR3) should be cleared before the system reset is released.

Note: The FSBL will always run if at least one of the reset flags that allows the FSBL to run is set.

7.4.3 FSBL Skip Function

To skip FSBL execution after a software reset release, RSTSR0, RSTSR1, and RSTSR3 should be cleared before executing software reset. If any flags other than RSTSR1.SWRF is set, FSBL may be executed.

To skip FSBL execution after a Deep Software Standby Reset release, RSTSR0, RSTSR1, and RSTSR3 should be cleared before the transition to Deep Software Standby mode. If any flags other than RSTSR0.DPSRSTF is set, FSBL may be executed.